

The Economy and the Global Water Cycle: an Integrated Assessment Model

Bruno Conte
UPF

Farid Farrokhi
Boston College

Hubert Massoni
U Bologna

bruno.conte@upf.edu

June 2026

Introduction

Economic & pop. growth → pressure on water resources (Rodell et al.2018, Nature)

- Globalization → water-intensive specialization (agriculture) in Brazil, China, India, US, ... (Debaere2014; Copeland et al.2022; Carleton et al.2024b, 2025)
- Concerning water use (w.r. to recharge) → billions exposed to future water insecurity (Mekonnen and Hoekstra2016, 2020, Science & Nature)

Introduction

Economic & pop. growth → pressure on water resources (Rodell et al.2018, Nature)

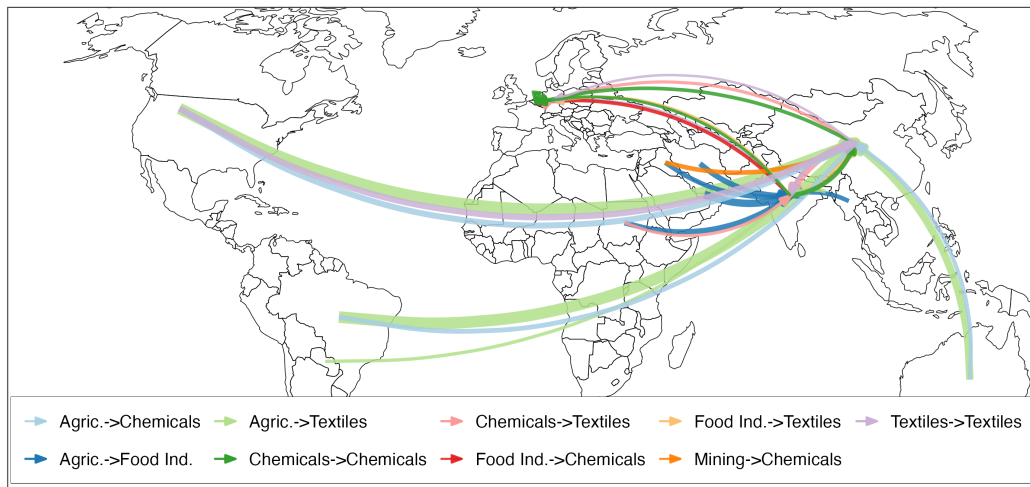
- Globalization → water-intensive specialization (agriculture) in Brazil, China, India, US, ... (Debaere2014; Copeland et al.2022; Carleton et al.2024b, 2025)
- Concerning water use (w.r. to recharge) → billions exposed to future water insecurity (Mekonnen and Hoekstra2016, 2020, Science & Nature)

This paper – these **local challenges reflect a global problem** due to spatial linkages

- Traded goods embody water footprints
- Water resources are linked through the global water cycle

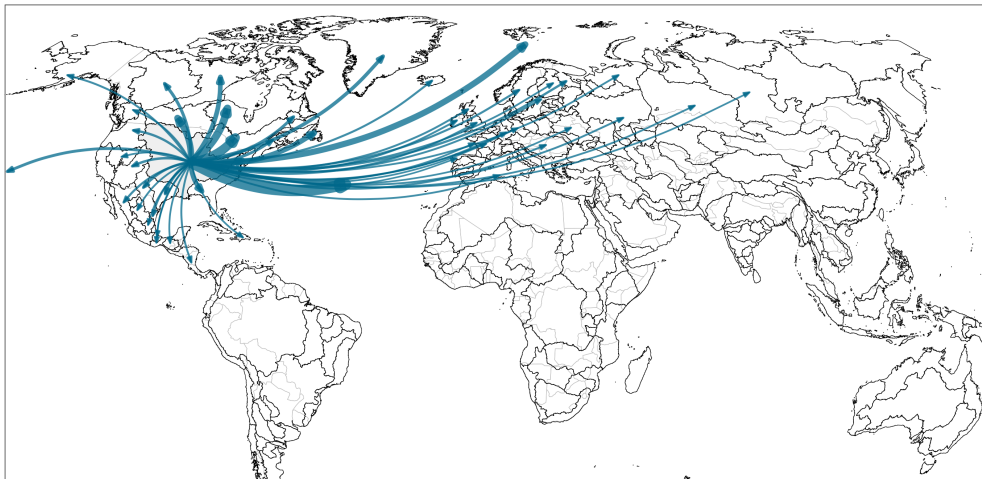
Globalized Economy → Globalized Water Footprint...

Fig. Water footprint (water m³/USD) of Germany's textile industry (GLORIA data, 2020)



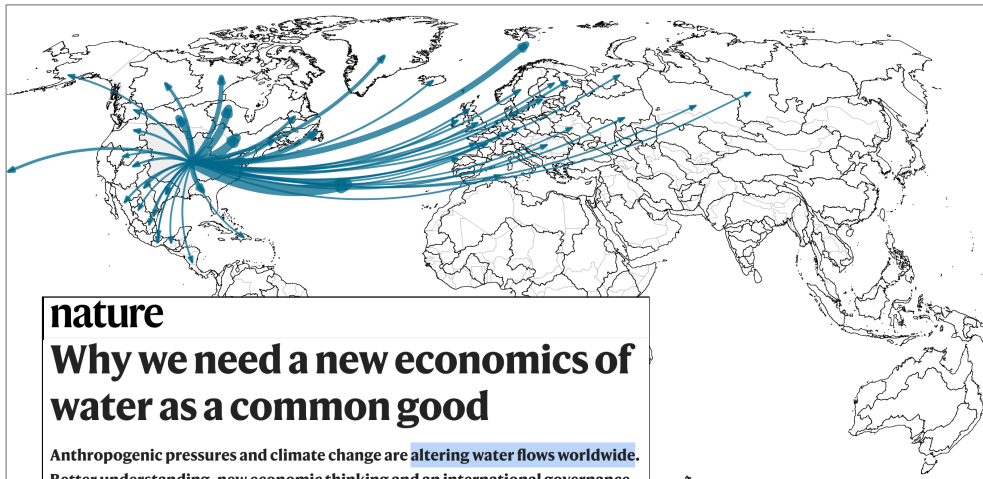
... Amplified by the Global Water Cycle

Fig. Evaporation-precipitation flows from the Mississippi Basin (RECON data; De Petrillo et al.2025a,b)



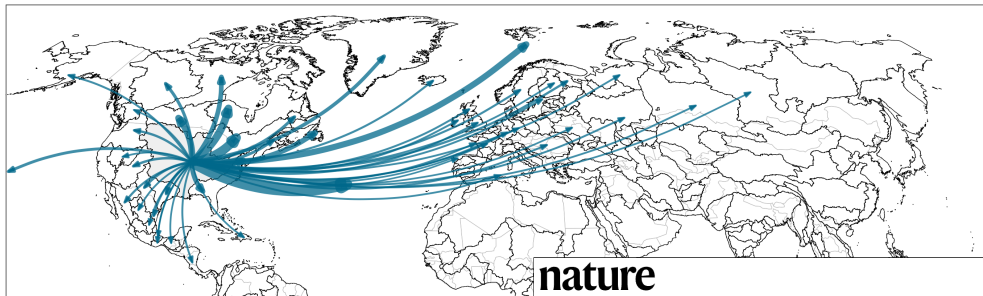
... Amplified by the Global Water Cycle

Fig. Evaporation-precipitation flows from the Mississippi Basin (RECON data; De Petrillo et al.2025a,b)



... Amplified by the Global Water Cycle

Fig. Evaporation-precipitation flows from the Mississippi Basin (RECON data; De Petrillo et al.2025a,b)



nature

**Why we need a new economic
water as a common good**

Anthropogenic pressures and climate change are altering water flows worldwide. Better understanding, new economic thinking and an international governance framework are needed to stave off catastrophe.

nature

**Carbon's social cost can't be
retrofitted to water**

What We Do (so far)

How does the co-evolution of economy & water cycle → water security and welfare?

- Can policies correct all (spatial and intertemporal) unpriced externalities?*

What We Do (so far)

How does the co-evolution of economy & water cycle → water security and welfare?

- Can policies correct all (spatial and intertemporal) unpriced externalities?*

1. Empirically show that the economy and water cycle are highly interconnected

- Climate-economy interaction → anthropogenic distortion of the water cycle

What We Do (so far)

How does the co-evolution of economy & water cycle → water security and welfare?

- Can policies correct all (spatial and intertemporal) unpriced externalities?*

1. Empirically show that the economy and water cycle are highly interconnected

- Climate-economy interaction → anthropogenic distortion of the water cycle

2. Develop a dynamic spatial model integrated with the global water cycle

- Common pool resource externalities → **globalized tragedy of the commons**

What We Do (so far)

How does the co-evolution of economy & water cycle → water security and welfare?

- Can policies correct all (spatial and intertemporal) unpriced externalities?*

1. Empirically show that the economy and water cycle are highly interconnected
 - Climate-economy interaction → anthropogenic distortion of the water cycle
2. Develop a dynamic spatial model integrated with the global water cycle
 - Common pool resource externalities → **globalized tragedy of the commons**
3. Derive an optimal water policy rule to internalize all unpriced externalities
 - Pigouvian price on water ↔ spatial "centrality" of water resources

Related Literature

policy contribution

Spatial-climate economics (Nordhaus1994; Desmet and Rossi-Hansberg2024) [more](#)

- Study the spatial climate-economy link + the role of trade, taxes, and policies
- Contribution: *the first integrated assessment model of the global water cycle*

Related Literature

policy contribution

Spatial-climate economics (Nordhaus1994; Desmet and Rossi-Hansberg2024) [more](#)

- Study the spatial climate-economy link + the role of trade, taxes, and policies
- Contribution: *the first integrated assessment model of the global water cycle*

Hydrology vs. water economics (e.g., Niazi et al.2024; Carleton et al.2025)

- Rich models that yet lack of full integration (or context specific) [water economics](#)
- Contribution: *bridge both fields, integrated framework for policy evaluation*

Related Literature policy contribution

Spatial-climate economics (Nordhaus1994; Desmet and Rossi-Hansberg2024) more

- Study the spatial climate-economy link + the role of trade, taxes, and policies
- Contribution: *the first integrated assessment model of the global water cycle*

Hydrology vs. water economics (e.g., Niazi et al.2024; Carleton et al.2025)

- Rich models that yet lack of full integration (or context specific) water economics
- Contribution: *bridge both fields, integrated framework for policy evaluation*

Resource econ. (Hotelling1931; Vanek1959; Solow1974; Davis and Weinstein2001)

- On the "limits to growth" & resource content of trade more
- *Study the intertemporal exploitation of spatially interconnected resources*

Roadmap

1. Data & Motivating Facts
2. A Dynamic Spatial Model of the Global Economy & Water Cycle
 - Integrated Model and Stylised Simulations
 - Optimal Water Policy for an Interconnected World
3. Next Steps: Linking Theory to Data + Counterfactuals + ...?

**Motivating facts: How interconnected
are the global water cycle & economy?**

Preamble: A Closed-System, Spatially Connected Water Cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} \kappa_j R_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} \delta_j R_j(t)}$$

Preamble: A Closed-System, Spatially Connected Water Cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - \underbrace{\underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)}}_{\substack{\uparrow \\ \text{Outflows (evaporation \& surface runoffs)}}} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} \kappa_j R_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} \delta_j R_j(t)}$$

Preamble: A Closed-System, Spatially Connected Water Cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \boxed{\underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} \kappa_j R_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} \delta_j R_j(t)}}$$

↑
Recharge (precipitation & surface runoffs)

Preamble: A Closed-System, Spatially Connected Water Cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} \kappa_j R_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} \delta_j R_j(t)}$$

 Atmospheric linkages

Preamble: A Closed-System, Spatially Connected Water Cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} \kappa_j R_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} \delta_j R_j(t)} \quad (1)$$

Economic water use $H_i(t)$ distorts eq. (1) – but in an ambiguous direction

- **Closed global system** $\rightarrow \sum_i R_i(t) = \sum_i R_i(t+1) \forall t$
- Withdraws **return to water system** (e.g., evotranspiration, surface runoff)

Preamble: A Closed-System, Spatially Connected Water Cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} \kappa_j R_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} \delta_j R_j(t)} \quad (1)$$

Economic water use $H_i(t)$ distorts eq. (1) – but in an ambiguous direction

- **Closed global system** $\rightarrow \sum_i R_i(t) = \sum_i R_i(t+1) \forall t$
- Withdraws **return to water system** (e.g., evotranspiration, surface runoff)

Next: empirical facts that corroborate these mechanisms (e.g., strength of $\{f_{ji}^{(E)}\}_{j,i}$)

Fact 1: Global Atmospheric Links Between Water Resources

RECON data: **atmospheric moisture flows**
($0.5^\circ \times 0.5^\circ$, De Petrillo et al.2025a,b) [details](#)

- Evap.→Precip. flows (m^3 , $\sim 2008-17$)

Fact 1: Global Atmospheric Links Between Water Resources

RECON data: **atmospheric moisture flows**
($0.5^\circ \times 0.5^\circ$, De Petrillo et al.2025a,b) [details](#)

- Evap.→Precip. flows (m^3 , ~2008-17)

Our focus: major land water basins (GAEZ)

- Basins $\equiv$ precipitation catchment areas
(no surface runoff b/w basins)

Fig. Land Water Basins from FAO GAEZ



Fact 1: Global Atmospheric Links Between Water Resources

RECON data: **atmospheric moisture flows**
($0.5^\circ \times 0.5^\circ$, De Petrillo et al.2025a,b) [details](#)

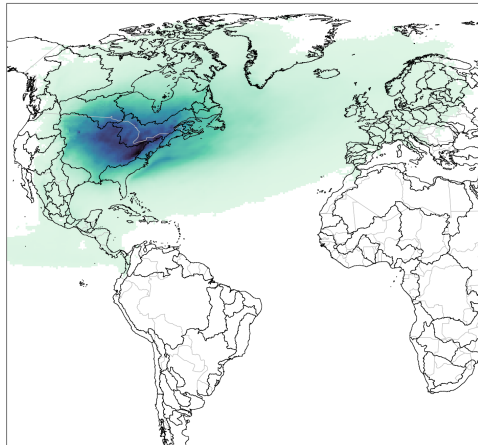
- Evap.→Precip. flows (m^3 , ~2008-17)

Our focus: major land water basins (GAEZ)

- Basins $\equiv$ precipitation catchment areas
(no surface runoff b/w basins)

Basin-level aggregation $(j, i) \equiv \sum_{g \in j} \sum_{g' \in i} f_{gg'}^{(E)}$

Fig. Land Water Basins from FAO GAEZ



Fact 1: Global Atmospheric Links Between Water Resources

RECON data: **atmospheric moisture flows**
($0.5^\circ \times 0.5^\circ$, De Petrillo et al.2025a,b) [details](#)

- Evap.→Precip. flows (m^3 , ~2008-17)

Our focus: major land water basins (GAEZ)

- Basins $\equiv$ precipitation catchment areas
(no surface runoff b/w basins)

Basin-level aggregation $(j, i) \equiv \sum_{g \in j} \sum_{g' \in i} f_{gg'}^{(E)}$

Fig. Land Water Basins from FAO GAEZ



Fact 1: Global Atmospheric Links Between Water Resources

RECON data: **atmospheric moisture flows**
($0.5^\circ \times 0.5^\circ$, De Petrillo et al.2025a,b) [details](#)

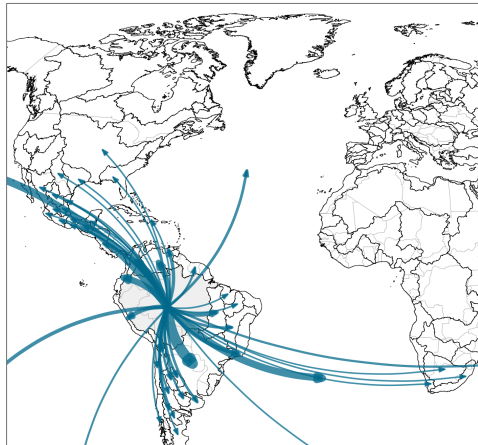
- Evap.→Precip. flows (m^3 , ~2008-17)

Our focus: major land water basins (GAEZ)

- Basins $\equiv$ precipitation catchment areas
(no surface runoff b/w basins)

Basin-level aggregation $(j, i) \equiv \sum_{g \in j} \sum_{g' \in i} f_{gg'}^{(E)}$

Fig. Land Water Basins from FAO GAEZ



Fact 1: Global Atmospheric Links Between Water Resources

RECON data: **atmospheric moisture flows**
($0.5^\circ \times 0.5^\circ$, De Petrillo et al.2025a,b) [details](#)

- Evap.→Precip. flows (m^3 , ~2008-17)

Our focus: major land water basins (GAEZ)

- Basins $\equiv$ precipitation catchment areas
(no surface runoff b/w basins)

Basin-level aggregation $(j, i) \equiv \sum_{g \in j} \sum_{g' \in i} f_{gg'}^{(E)}$

Fig. Land Water Basins from FAO GAEZ



Fact 1: Global Atmospheric Links Between Water Resources

Table 1: Water "import shares" per origin basin type (%) [maps](#) [export shares](#)

	Own basin	Other basins (domestic)	Other basins (foreign)	Oceans
Min.	0.65	0.09	0.17	5.82
Q1	2.41	3.68	7.94	52.99
Median	4.81	8.62	17.71	68.72
Mean	7.68	11.12	20.04	63.99
Q3	9.61	16.59	28.58	79.09
Max.	48.66	50.93	71.27	98.18

Fact 1: Global Atmospheric Links Between Water Resources

Table 1: Water "import shares" per origin basin type (%) [maps](#) [export shares](#)

	Own basin	Other basins (domestic)	Other basins (foreign)	Oceans
Min.	0.65	0.09	0.17	5.82
Q1	2.41	3.68	7.94	52.99
Median	4.81	8.62	17.71	68.72
Mean	7.68	11.12	20.04	63.99
Q3	9.61	16.59	28.58	79.09
Max.	48.66	50.93	71.27	98.18

Median basin ~ 25% from other land basins

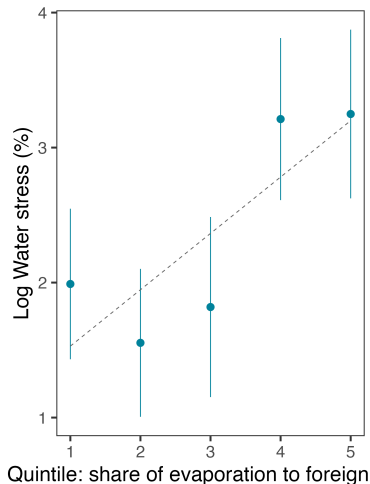
Fact 2: Most water-stressed countries $\leftrightarrow$ top "atm. exporters"

Water stress $\equiv$ water withdraws/endowments FAO data

Top water-stressed countries:

- Also the top "atmospheric water exporters"
- I.e., most central countries in terms of "propagation" of local water depletion

Fig. Water exporters vs. water stress



Fact 2: Most water-stressed countries $\leftrightarrow$ top "atm. exporters"

Water stress $\equiv$ water withdraws/endowments [FAO data](#)

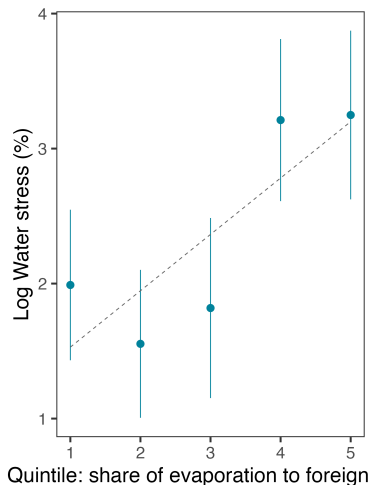
Top water-stressed countries:

- Also the top "atmospheric water exporters"
- I.e., most central countries in terms of "propagation" of local water depletion

GLORIAMRIO (164 countries, 46 sectors, 1995-2020)

- Water m^3/USD $\rightarrow$ water content of trade [details](#)
- 2000-20: $\uparrow$ specialization w.r. to traded water

Fig. Water exporters vs. water stress



Motivating Facts: Taking Stock

Global economy & water cycle – highly interconnected networks

- Local water exploitation → potential global consequences (via feedback loops)
- Current water challenges: cross-sectional evidence of a **slow-moving process**

How will the economy & water co-evolve, and where it will lead is unknown

Next: an integrated model that incorporates such interconnections

- Optimal water policy that corrects all water-related externalities

Theory: A Dynamic Spatial Model of the Global Economy & Water Cycle

Dynamic Spatial Model – Overview

Economic module: trade model with labor and water as inputs

- Subject to policies: tariffs, subsidies, water prices
- Water use → higher extraction/prod. costs → shifts in trade & production

Water cycle module: closed system of water basins whose stock are

- Depleted by evaporation (and water use), replenished by precipitation
- Economic water use returns to the (closed) system (evotranspiration, runoffs)

Dynamic Spatial Model – Overview

Economic module: trade model with labor and water as inputs

- Subject to policies: tariffs, subsidies, water prices
- Water use → higher extraction/prod. costs → shifts in trade & production

Water cycle module: closed system of water basins whose stock are

- Depleted by evaporation (and water use), replenished by precipitation
- Economic water use returns to the (closed) system (evotranspiration, runoffs)

Water-Economy ↻ Feedback → Globalized Tragedy of the Commons

(i.e., local extraction imposes unpriced costs on the rest of the world)

Dynamic Spatial Model – Environment

Geography of N countries i, j ; fixed labor $\bar{L}_i$ and endogenous water stocks $R_i(t)$

- $K + 1$ industries ($k = 0$: water extraction; $k \geq 1$: final goods), time $t = 0, 1, \dots$

Dynamic Spatial Model – Environment

Geography of N countries i, j ; fixed labor $\bar{L}_i$ and endogenous water stocks $R_i(t)$

- $K + 1$ industries ($k = 0$: water extraction; $k \geq 1$: final goods), time $t = 0, 1, \dots$

Policy instruments create a wedge b/w producer (P) and consumer prices ($\tilde{P}$):

$$\tilde{P}_{ji,k} = \frac{(1 + t_{ji,k})}{(1 + s_{j,k})(1 + x_{ji,k})} \times P_{ji,k}$$

The diagram illustrates the relationship between producer and consumer prices. The equation $\tilde{P}_{ji,k} = \frac{(1 + t_{ji,k})}{(1 + s_{j,k})(1 + x_{ji,k})} \times P_{ji,k}$ is shown. Three blue arrows point to the terms in the equation: an arrow from 'Import Tariff' points to $t_{ji,k}$, an arrow from 'Production Subsidy' points to $s_{j,k}$, and an arrow from 'Export Subsidy' points to $x_{ji,k}$.

Dynamic Spatial Model – Environment

Geography of N countries i, j ; fixed labor $\bar{L}_i$ and endogenous water stocks $R_i(t)$

- $K + 1$ industries ($k = 0$: water extraction; $k \geq 1$: final goods), time $t = 0, 1, \dots$

Policy instruments create a wedge b/w producer (P) and consumer prices ($\tilde{P}$):

$$\tilde{P}_{ji,k} = \frac{(1 + t_{ji,k})}{(1 + s_{j,k})(1 + x_{ji,k})} \times P_{ji,k}$$

The diagram illustrates the relationship between producer and consumer prices. The equation $\tilde{P}_{ji,k} = \frac{(1 + t_{ji,k})}{(1 + s_{j,k})(1 + x_{ji,k})} \times P_{ji,k}$ is shown. Three blue arrows point to the terms in the equation: an arrow from 'Import Tariff' points to $t_{ji,k}$, an arrow from 'Production Subsidy' points to $s_{j,k}$, and an arrow from 'Export Subsidy' points to $x_{ji,k}$.

Tax revenues T_i rebated to households $\rightarrow$ balanced budget: $I_i = w_i \bar{L}_i + T_i$

- Countries can also resort to **water pricing** $\tau_{i,k}$ (defined later)

Dynamic Spatial Model – Demand & Supply

Demand: $W_i = V_i(I_i, \tilde{P}_i)$; Marshallian demand (Roy's identity) $C_{ji,k} = D_{ji,k}(I_i, \tilde{P}_i)$; e.g.,

$$W_i = \frac{I_i}{\tilde{P}_i}, \quad \tilde{P}_i = \prod_k \left[\sum_n b_{ni,k} \tilde{P}_{ni,k}^{1-\sigma_k} \right]^{\frac{\beta_{i,k}}{1-\sigma_k}}, \quad D_{ji,k} = \frac{b_{ji,k} \tilde{P}_{ji,k}^{-\sigma_k}}{\sum_n b_{ni,k} \tilde{P}_{ni,k}^{1-\sigma_k}} \beta_{i,k} I_i \quad (\text{CD-CES})$$

Dynamic Spatial Model – Demand & Supply

Demand: $W_i = V_i(I_i, \tilde{P}_i)$; Marshallian demand (Roy's identity) $C_{ji,k} = D_{ji,k}(I_i, \tilde{P}_i)$; e.g.,

$$W_i = \frac{I_i}{\tilde{P}_i}, \quad \tilde{P}_i = \prod_k \left[\sum_n b_{ni,k} \tilde{P}_{ni,k}^{1-\sigma_k} \right]^{\frac{\beta_{i,k}}{1-\sigma_k}}, \quad D_{ji,k} = \frac{b_{ji,k} \tilde{P}_{ji,k}^{-\sigma_k}}{\sum_n b_{ni,k} \tilde{P}_{ni,k}^{1-\sigma_k}} \beta_{i,k} I_i \quad (\text{CD-CES})$$

Supply: Final goods ($k \geq 1$) with CRS production $(L_{j,k}, H_{j,k})$; producer price:

$$P_{ji,k} = \bar{d}_{ji,k} c_{j,k}(w_j, \tilde{r}_{j,k}), \quad \text{where}$$

$\tilde{r}_{j,k} = r_j + \tau_{j,k} \equiv$ effective water price (inclusive of **water pricing** $\tau_{j,k}$)

Dynamic Spatial Model – Water Extraction & Factor Demands

Water extraction ($k = 0$) uses labor as $Q_{j,0} = L_{j,0} / G_j(R_j) \rightarrow$

$$r_j = w_j G_j(R_j)$$

where $G'_j < 0 \rightarrow$ extraction costs increase as resources deplete

Dynamic Spatial Model – Water Extraction & Factor Demands

Water extraction ($k = 0$) uses labor as $Q_{j,0} = L_{j,0} / G_j(R_j) \rightarrow$

$$r_j = w_j G_j(R_j)$$

where $G'_j < 0 \rightarrow$ extraction costs increase as resources deplete

Factor demands: $L_{j,k} = \frac{\partial c_{j,k}}{\partial w_j} Q_{j,k}$, $H_{j,k} = \frac{\partial c_{j,k}}{\partial r_{j,k}} Q_{j,k}$ (Shephard's lemma)

- $H_{j,k}$ rises when R_j is abundant (low r_j) and falls as the resource depletes

Myopic agents $\rightarrow$ **Common Pool Resource Externality** (spatially and intertemporally)

(i.e., firms take r_j as given, not internalizing that withdrawals lower R_j and affect water stocks (and prices) in all connected basins)

Water Cycle – Basin Geography & Natural Hydrology

Global water cycle as a **closed system** of land basins $\mathcal{B} = \{1, \dots, B\}$ + ocean $\mathcal{O} = \{0\}$

- Oceans treated as a stable reservoir (i.e., $E_0 = \bar{E}_0$, $S_0 = \bar{S}_0$ exogenous)

Without economic activity, water stocks evolve as

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} E_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} R_j(t)}$$

Water Cycle – Basin Geography & Natural Hydrology

Global water cycle as a **closed system** of land basins $\mathcal{B} = \{1, \dots, B\}$ + ocean $\mathcal{O} = \{0\}$

- Oceans treated as a stable reservoir (i.e., $E_0 = \bar{E}_0$, $S_0 = \bar{S}_0$ exogenous)

Without economic activity, water stocks evolve as

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} E_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} R_j(t)}$$

Stacking land basins $i \geq 1 \rightarrow$ law of motion:

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t)$$

$$\mathbf{\Gamma} \equiv \mathbf{F}_{OB}^{(E)} \bar{\mathbf{E}}_0 + \mathbf{F}_{OB}^{(S)} \bar{\mathbf{S}}_0 \quad (\text{exogenous ocean inflows})$$

$$\mathbf{\Phi} \equiv \mathbf{I} - \mathbf{\kappa} - \mathbf{\delta} + \mathbf{F}_{BB}^{(E)} \mathbf{\kappa} \quad (\text{net retention: losses + atm. recycling})$$

Water Cycle: Economic Withdrawals & Consumptive Use

Now add economic activity with **consumptive-use share** $\theta_{ik} \in (0, 1)$:

- Fraction of withdrawal lost via evaporation; remainder returns as runoff

Basin-level aggregates:

$$H_i = \sum_k H_{ik}, \quad \theta_i(t) = \frac{\sum_k \theta_{ik} H_{ik}}{\sum_k H_{ik}} \quad (\text{time-varying with sectoral mix})$$

Modified evapotranspiration (natural + economic):

$$E_B(t) = \kappa R_B(t) + \theta(t) H(t)$$

Economic withdrawals $\uparrow$ evapotranspiration, shifting precipitation $\forall$ basins

Water Cycle: Updated Law of Motion & Steady State

Law of motion with economic activity:

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t) + \mathbf{\Psi}(t) \mathbf{H}(t), \quad \mathbf{\Psi}(t) \equiv (\mathbf{F}_{BB}^{(E)} - \mathbf{I}) \boldsymbol{\theta}(t)$$

$\mathbf{\Psi}(t) \equiv$ *net economic retention*, with an ambiguous sign

Steady state with economic activity (fixed-point in $\mathbf{R}_B^{(ss)}$):

$$\mathbf{R}_B^{(ss)} = (\mathbf{I} - \mathbf{\Phi})^{-1} (\mathbf{\Gamma} + \mathbf{\Psi}^{(ss)} \mathbf{H}(\mathbf{R}_B^{(ss)}))$$

Closed system: $\sum_i R_i(t+1) = \sum_i R_i(t)$ – fixed global water stock

General Equilibrium

Static equilibrium – wages $\{w_i\}$ simultaneously clear:

$$\text{(Goods)} \quad Q_{i,k} = \sum_n \bar{d}_{in,k} C_{in,k} \quad \text{(Water)} \quad \frac{L_{i,0}}{G_i(R_i)} = \sum_{k=1}^K H_{i,k} \quad \text{(Labor)} \quad \bar{L}_i - L_{i,0} = \sum_{k=1}^K L_{i,k}$$

Dynamic equilibrium: sequence of static equilibria $\{w_i(t)\}_t$ with $R_i(0)$ given

- Stocks $\{R_i(t)\}_t$ evolving per the LOM above (state variable)

Steady-state equilibrium: dynamic equilibrium satisfying the $R_B^{(ss)}$ fixed-point

solving algor.

simulations

Optimal Water Policy for and Interconnected World

Optimal Policy – Problem & Water Stock Linearization

Optimal policy $\mathbb{I}_i \equiv \{\mathbf{t}_i, \mathbf{x}_i, \mathbf{s}_i, \tau_i\}$ + transfers Δ_i ($\sum_i \Delta_i = 0$) solves

$$\max_{\mathbb{I}} \mathcal{W}(W_1(\mathbb{I}), \dots, W_N(\mathbb{I})) \quad \text{s.to steady-state GE relationships}$$

Optimal Policy – Problem & Water Stock Linearization

Optimal policy $\mathbb{I}_i \equiv \{\mathbf{t}_i, \mathbf{x}_i, \mathbf{s}_i, \tau_i\} + \text{transfers } \Delta_i \text{ } (\sum_i \Delta_i = 0)$ solves

$$\max_{\mathbb{I}} \mathcal{W}(W_1(\mathbb{I}), \dots, W_N(\mathbb{I})) \quad \text{s.to steady-state GE relationships}$$

Re-write steady-state water stocks (from the LOM fixed point) as:

$$\mathbf{R}_B^{(ss)} = \mathbf{Z} + \mathbf{\Lambda}^{(ss)} \mathbf{H}^{(ss)}, \quad \mathbf{Z} \equiv (\mathbf{I} - \mathbf{\Phi})^{-1} \mathbf{\Gamma}, \quad \mathbf{\Lambda}^{(ss)} \equiv (\mathbf{I} - \mathbf{\Phi})^{-1} \mathbf{\Psi}^{(ss)}$$

$$R_i = Z_i + \sum_j \sum_k \lambda_{ji,k} H_{j,k} \quad [\text{country level}]$$

$\lambda_{ji,k}$: hydrological "impact factor" of withdrawals at (j, k) on steady-state R_i

Optimal Water Policy – Tax Rates derivation

Non-water taxes: $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$

- From a global perspective, distorting goods prices is always inefficient

Optimal Water Policy – Tax Rates derivation

Non-water taxes: $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$

- From a global perspective, distorting goods prices is always inefficient

Optimal water tax (Pigouvian + spatial linkages):

$$\tau_{i,k} = \tilde{r}_{i,k} - r_i = \sum_n \frac{G'_n(\cdot)}{G_n(\cdot)} r_n \sum_g \lambda_{in,g} H_{n,g}$$

- $G'_n/G_n < 0$: *extraction externality* – depletion → higher future unit costs
- $\lambda_{in,g}$: *spatial weights* – hydrological linkages propagate the externality globally

Optimal Policy – International Transfers

Optimal income shares achieving any distributional objective:

$$\pi_i^* = \frac{\frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial V_i}{\partial E_i} I_i}{\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} I_n}, \quad \Delta_i^* = \pi_i^* \sum_n I_n - I_i$$

Optimal Policy – International Transfers

Optimal income shares achieving any distributional objective:

$$\pi_i^* = \frac{\frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial V_i}{\partial E_i} I_i}{\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} I_n}, \quad \Delta_i^* = \pi_i^* \sum_n I_n - I_i$$

Special case: log SWF $\mathcal{W} = \sum_i \omega_i \ln W_i$ + homothetic demand

- $\pi_i^* = \omega_i \Rightarrow$ efficient frontier traced by varying $\{\omega_i\}$

Optimal Policy – International Transfers

Optimal income shares achieving any distributional objective:

$$\pi_i^* = \frac{\frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial V_i}{\partial E_i} I_i}{\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} I_n}, \quad \Delta_i^* = \pi_i^* \sum_n I_n - I_i$$

Special case: log SWF $\mathcal{W} = \sum_i \omega_i \ln W_i$ + homothetic demand

- $\pi_i^* = \omega_i \Rightarrow$ efficient frontier traced by varying $\{\omega_i\}$

Optimal policy separates efficiency from distribution: water taxes correct the externality; transfers achieve the distributional goal

Final Remarks & Next Steps

Final Remarks & Next Steps

Economic activity distorts the global water cycle through spatial linkages

- Slow-moving but empirically measurable process

A dynamic spatial model integrating both networks

- Common-pool externalities → *globalized tragedy of the commons*
- Corrected by a **basin-specific Pigouvian water tax** ~ Social Cost of Water

Next steps: link to global data → quantify welfare costs of water externalities

- Simulate optimal policies for the 21st century
- Model extension (e.g., full-fledged dynamics, non-homothetic preferences, ...)

Thank you

bruno.conte@upf.edu

I acknowledge funding from the EU's Horizon Europe research and innovation programme (Marie Skłodowska-Curie grant agreement No 101146979-SPEED), the EIEF 2023 grant "International Trade, Globalization, and the Environment", the Ramón Areces Social Sciences grant "Trade Policy and Water Scarcity Around the World", and the BSE Seed Grant SG2026-05 "Water Scarcity (and Policy) in the Globalized Economy".

References I

- Blakeslee, David, Aaditya Dar, Ram Fishman, Samreen Malik, Heitor S Pellegrina, and Karan Singh Bagavathinathan**, “Irrigation and the spatial pattern of local economic development in India,” *Journal of Development Economics*, 2023, 161, 102997.
- Bourany, Thomas**, “Climate Change, Inequality, and the Optimal Climate Policy,” *Working Paper*, 2025.
- , “The Optimal Design of Climate Agreements: Inequality, Trade, and Incentives for Climate Policy,” *Working Paper*, 2025.
- and **Jordan Rosenthal-Kay**, “The winners and losers of climate policies: a sufficient statistics approach,” *Working Paper*, 2025.
- Burke, Marshall and Kyle Emerick**, “Adaptation to climate change: Evidence from US agriculture,” *American Economic Journal: Economic Policy*, 2016, 8 (3), 106–140.
- Carleton, Tamma, Esther Duflo, B Kelsey Jack, and Guglielmo Zappalà**, “Adaptation to climate change,” in “Handbook of the Economics of Climate Change,” Vol. 1, Elsevier, 2024, pp. 143–248.
- , **Levi Crews, and Ishan Nath**, “Is the world running out of fresh water?,” in “AEA Papers and Proceedings,” Vol. 114 American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203 2024, pp. 31–35.
- , —, and —, “Agriculture, trade, and the spatial efficiency of global water use,” *Working Paper*, 2025.
- Conte, Bruno, Klaus Desmet, and Esteban Rossi-Hansberg**, “On the Geographic Implications of Carbon Taxes,” *The Economic Journal*, 04 2025, p. ueaf027.
- , —, **Dávid Krisztián Nagy, and Esteban Rossi-Hansberg**, “Local sectoral specialization in a warming world,” *Journal of Economic Geography*, 2021, 21 (4), 493–530.
- Copeland, Brian R, Joseph S Shapiro, and M Scott Taylor**, “Globalization and the environment,” in “Handbook of International Economics,” Vol. 5, Elsevier, 2022, pp. 61–146.
- Cruz, José-Luis and Esteban Rossi-Hansberg**, “The economic geography of global warming,” *Review of Economic Studies*, 2024, 91 (2), 899–939.

References II

- Davis, Donald R and David E Weinstein**, “An account of global factor trade,” *American Economic Review*, 2001, 91 (5), 1423–1453.
- Debaere, Peter**, “The global economics of water: is water a source of comparative advantage?,” *American Economic Journal: Applied Economics*, 2014, 6 (2), 32–48.
- and **Tianshu Li**, “The effects of water markets: Evidence from the Rio Grande,” *Advances in Water Resources*, 2020, 145, 103700.
- Desmet, Klaus and Esteban Rossi-Hansberg**, “On the spatial economic impact of global warming,” *Journal of Urban Economics*, 2015, 88, 16–37.
- and —, “Climate change economics over time and space,” *Annual Review of Economics*, 2024, 16.
- Farrokhi, Farid**, “Global sourcing in oil markets,” *Journal of International Economics*, 2020, 125, 103323.
- , **Ahmad Lashkaripour**, and **Homa Taheri**, “A Framework for Integrating Climate Goals into Trade Agreements,” *Working Paper*, 2025.
- , **Elliot Kang**, **Heitor S Pellegrina**, and **Sebastian Sotelo**, “Deforestation: A global and dynamic perspective,” Technical Report, National Bureau of Economic Research 2025.
- Fishman, Ram, Naresh Devineni, and Swaminathan Raman**, “Can improved agricultural water use efficiency save India’s groundwater?,” *Environmental Research Letters*, 2015, 10 (8), 084022.
- Haseeb, Muhammad**, “Resource Scarcity and Cooperation,” *Working Paper*, 2024.
- Hassler, John and Per Krusell**, “Economics and climate change: integrated assessment in a multi-region world,” *Journal of the European Economic Association*, 2012, 10 (5), 974–1000.
- , —, and **Conny Olovsson**, “Finite resources and the world economy,” *Journal of International Economics*, 2022, 136, 103592.
- Hornbeck, Richard and Pinar Keskin**, “The historically evolving impact of the ogallala aquifer: Agricultural adaptation to groundwater and drought,” *American Economic Journal: Applied Economics*, 2014, 6 (1), 190–219.

References III

- Hotelling, Harold**, “The economics of exhaustible resources,” *Journal of political Economy*, 1931, 39 (2), 137–175.
- Kang, Elliot**, “Global Fisheries: Quantifying the Externalities from Open Access,” *Job Market Paper, University of Michigan*, 2024, 61.
- Krusell, Per and Anthony A Smith**, “Climate Change Around the World,” *NBER Working Paper*, 2022, (w30338).
- Mazzucato, Mariana, Ngozi Okonjo-Iweala, Johan Rockström, and Tharman Shanmugaratnam**, “The economics of water: valuing the hydrological cycle as a global common good,” 2024.
- Mekonnen, Mesfin M and Arjen Y Hoekstra**, “Four billion people facing severe water scarcity,” *Science advances*, 2016, 2 (2), e1500323.
- and —, “Blue water footprint linked to national consumption and international trade is unsustainable,” *Nature Food*, 2020, 1 (12), 792–800.
- Niazi, Hassan, Thomas B Wild, Sean WD Turner, Neal T Graham, Mohamad Hejazi, Siwa Msangi, Son Kim, Jonathan R Lamontagne, and Mengqi Zhao**, “Global peak water limit of future groundwater withdrawals,” *Nature Sustainability*, 2024, 7 (4), 413–422.
- Nordhaus, William**, *Managing the global commons: the economics of climate change*, Vol. 31, MIT press Cambridge, MA, 1994.
- Olmstead, Sheila M**, “The economics of managing scarce water resources,” *Review of Environmental Economics and policy*, 2010.
- Petrillo, Elena De, Luca Monaco, Marta Tuninetti, Arie Staal, and Francesco Laio**, “Cell-scale atmospheric moisture flows dataset reconciled with ERA5 reanalysis,” *Scientific data*, 2025, 12 (1), 629.
- , **Simon Felix Fahrländer, Marta Tuninetti, Lauren Seaby Andersen, Luca Monaco, Luca Ridolfi, and Francesco Laio**, “Reconciling tracked atmospheric water flows to close the global freshwater cycle,” *Communications Earth & Environment*, 2025, 6 (1), 347.

References IV

- Pindyck, Robert S**, “The optimal exploration and production of nonrenewable resources,” *Journal of political economy*, 1978, 86 (5), 841–861.
- Rodell, Matthew, Jay S Famiglietti, David N Wiese, JT Reager, Hiroko K Beaudoin, Felix W Landerer, and M-H Lo**, “Emerging trends in global freshwater availability,” *Nature*, 2018, 557 (7707), 651–659.
- Sekhri, Sheetal**, “Wells, water, and welfare: the impact of access to groundwater on rural poverty and conflict,” *American Economic Journal: Applied Economics*, 2014, 6 (3), 76–102.
- Solow, Robert M**, “The Economics of Resources or the Resources of Economics,” *The American Economic Review*, 1974, 64 (2), 1–14.
- Stiglitz, Joseph**, “Growth with Exhaustible Natural Resources: Efficient and Optimal Growth Paths,” *The Review of Economic Studies*, 1974, 41 (5), 123–137.
- Vanek, Jaroslav**, “The natural resource content of foreign trade, 1870–1955, and the relative abundance of natural resources in the United States,” *The Review of Economics and Statistics*, 1959, pp. 146–153.

Appendix

Policy Contribution: Details [Back](#)

Policy-relevant: addresses calls for global water governance (Mazzucato et al.2024) and HORIZON EU's environmental research priorities

Envisioned legacy: inspire new research on

- **Social cost of water:** challenges the paradigm of an inherently local SCW (Olmstead2010) → first calculation of shadow prices internalizing **full spatial & intertemporal externalities** of local water use
- **Political economy** of shared (water) resources
- Water and the **green transition:** energy-intensive technologies, AI, ...

Spatial-climate economics: spatial economy–climate (Nordhaus1994; Hassler and Krusell2012; Krusell and Smith2022; Desmet and Rossi-Hansberg2015, 2024)

- Trade (Conte et al.2021), green subsidies (Cruz and Rossi-Hansberg2024), CO₂ taxes (Conte et al.2025), policy coordination (Farrokhi et al.2025a)
- Distributional effects (Bourany2025a,b; Bourany and Rosenthal-Kay2025)

Gap: no study has focused on water as an outcome of the economy–climate system

- *This paper:* first integrated model of the global economy & global water cycle

Hydrology: rich models projecting future water challenges, but limited flexibility for policy exercises (e.g., Niazi et al.2024)

Water economics: mostly context-specific settings

- India (e.g., Fishman et al.2015; Sekhri2014; Blakeslee et al.2023); Pakistan (Haseeb2024); US (e.g., Hornbeck and Keskin2014; Debaere and Li2020)

Gap: lack of full integration (e.g., exogenous recharge; Carleton et al.2025)

- *This paper:* global, integrated & transparent framework **bridging both fields**

"Limits to growth": intertemporal resource depletion (Solow1974; Stiglitz1974; Pindyck1978); resource content of trade (Vanek1959; Davis and Weinstein2001; Debaere2014)

Modern revival – sustainable exploitation in spatial models:

- Forests (Farrokhi et al.2025b), fisheries (Kang2024), oil (Farrokhi2020; Hassler et al.2022)

Gap: the modern revival has not yet extended to water

- *This paper*: intertemporal exploitation of **spatially interconnected** water resources

RECON (De Petrillo et al.2025a,b): bilateral atmospheric moisture flows between every pair of 0.5° grid cells worldwide

- **Methodology:** UTrack Lagrangian moisture tracking $\rightarrow$ Iterative Proportional Fitting (IPF) reconciliation with ERA5 reanalysis
- **Coverage:** global, $0.5^\circ \times 0.5^\circ$; average year based on **2008–2017**
- **Content:** $f_{gg'}^{(E)}$ (m^3) = atmospheric moisture evaporated at cell g and precipitated at cell g'

Basin-level aggregation $\rightarrow$ moisture matrix $F_{BB}^{(E)}$ used in the model's law of motion:

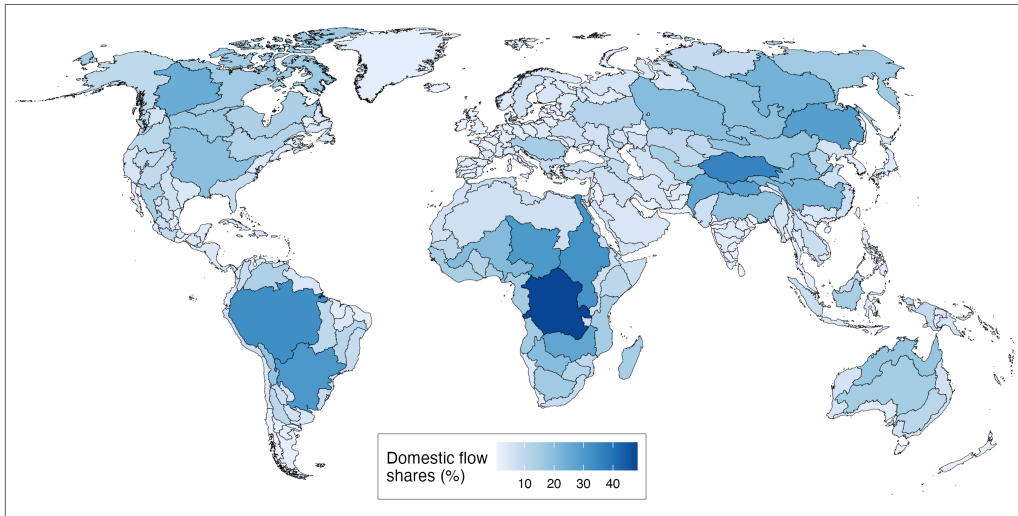
$$f_{ji}^{(E)} = \sum_{g \in j} \sum_{g' \in i} f_{gg'}^{(E)}$$

Country-level aggregation $\rightarrow$ “foreign water use” regressor in Fact 2:

$$f_{c'c}^{(E)} = \sum_{g \in c'} \sum_{g' \in c} f_{gg'}^{(E)}, \quad \Delta \text{Foreign water use}_c = \sum_{c' \neq c} f_{c'c}^{(E)} \Delta \text{Water use}_{c'}$$

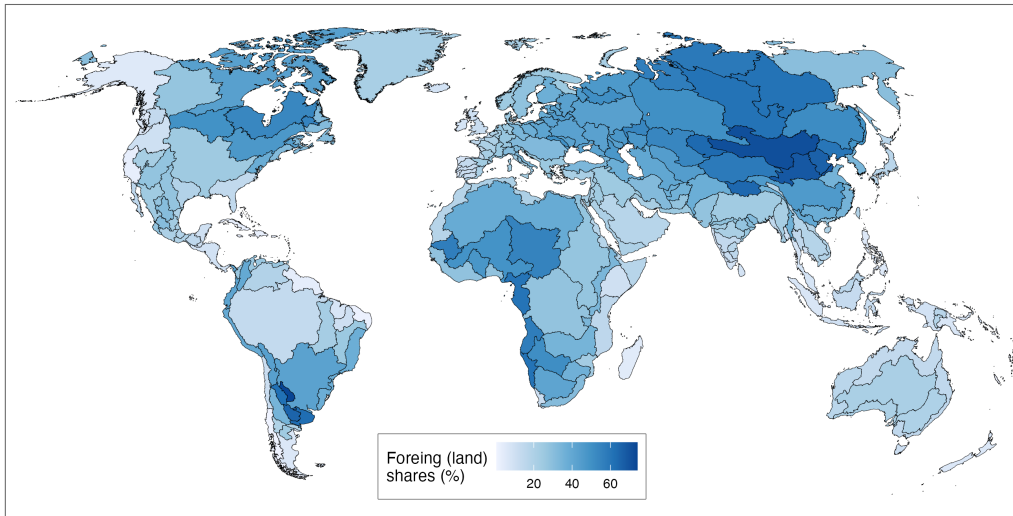
Atmospheric Moisture Shares: Maps [Back](#)

Fig. Share of atmospheric "domestic" flows



Atmospheric Moisture Shares: Maps [Back](#)

Fig. Share of atmospheric "foreign flows" from other land basins



Atmospheric Moisture Shares: Maps [Back](#)

Fig. Share of atmospheric "foreign flows" from the oceans

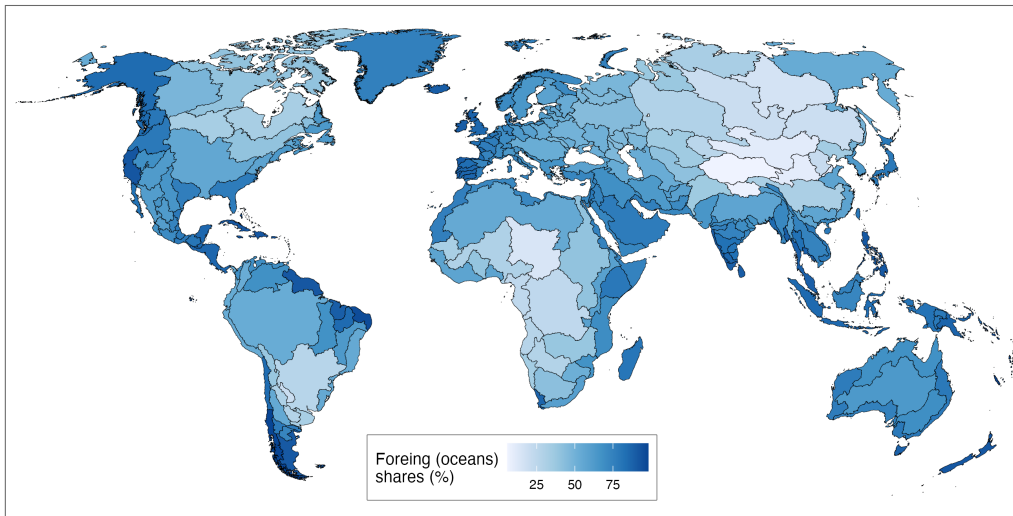


Table 2: Water "export shares" per destination basin type (%)

	Own basin	Other basins (domestic)	Other basins (foreign)	Oceans
Min.	0.77	0.06	1.39	1.60
Q1	3.01	5.59	13.63	19.43
Median	6.59	13.34	28.14	34.34
Mean	11.04	20.37	36.20	37.58
Q3	13.04	32.62	57.22	53.91
Max.	74.58	80.21	91.43	94.26

ERA5 (ECMWF Reanalysis v5): accessed via Copernicus Climate Data Store

- **Variables:** monthly evaporation, precipitation, and 2m temperature
- **Spatial resolution:** $0.25^{\circ} \times 0.25^{\circ}$ (~ 31 km at equator); global
- **Temporal span:** 1940 to present (monthly averages)

Processing: aggregate from grid-cell $\times$ month $\rightarrow$ country $\times$ year

- Area-weighted average of 0.25° cells within each country's borders
- Sum (evaporation, precipitation) or average (temperature) over months within year

FAO-AQUASTAT – main water use variable:

- Country-level freshwater withdrawals by sector (agricultural, municipal, industrial)
- 193 countries, ~1960–present (irregular frequency by country)

World Bank Development Indicators – proxy variables:

- Agricultural value added (constant USD; % of GDP)
- Agricultural land area (km²; % of total land area)
- ~200 countries, 1960–present

Country Trends: Construction [Back](#)

Measuring long-run Δ : country-level trends estimated from the panel (not endpoint differences)

- Less sensitive to year-specific outliers and business-cycle fluctuations

For each outcome o (e.g., water use), estimate country-specific linear time trends:

$$\ln y_{ct}^o = \sum_{c'} \gamma_c^o \cdot (\mathbf{1}\{c = c'\} \times \text{Year}_t) + \varepsilon_{ct}$$

$\hat{\gamma}_c^o \equiv$ estimated average annual growth rate (%) of outcome o in country c

- $\hat{\gamma}_c$ used as Δy_c^o in the cross-country regressions

Fact 2: Economic Activity Explains Δ Global Water Cycle [Back](#)

Table 3: Regression results – global water cycle and agricultural VA

	Δ Evaporation			Δ Precipitation		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Agric. VA	0.6629*** (0.0582)	0.6820*** (0.0578)		0.6975*** (0.0685)		0.6085*** (0.0701)
Δ Temperature		0.0278** (0.0114)	0.0096 (0.0156)	0.0032 (0.0135)	-0.0155 (0.0173)	0.0147 (0.0134)
Δ Foreign agric. VA						0.0143*** (0.0039)
Observations	156	156	156	156	156	156
R ²	0.45708	0.47731	0.00243	0.40703	0.00519	0.45569

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Fact 2: Economic Activity Explains Δ Global Water Cycle [Back](#)

Table 4: Regression results – global water cycle and agricultural land use

	Δ Evaporation			Δ Precipitation		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Agric. land	0.6886*** (0.0388)	0.6955*** (0.0383)		0.6828*** (0.0515)		0.6210*** (0.0545)
Δ Temperature		0.0216** (0.0088)	0.0096 (0.0156)	-0.0038 (0.0119)	-0.0155 (0.0173)	0.0051 (0.0120)
Δ Foreign agric. land						0.0208*** (0.0071)
Observations	156	156	156	156	156	156
R ²	0.67204	0.68432	0.00243	0.53678	0.00519	0.56158

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

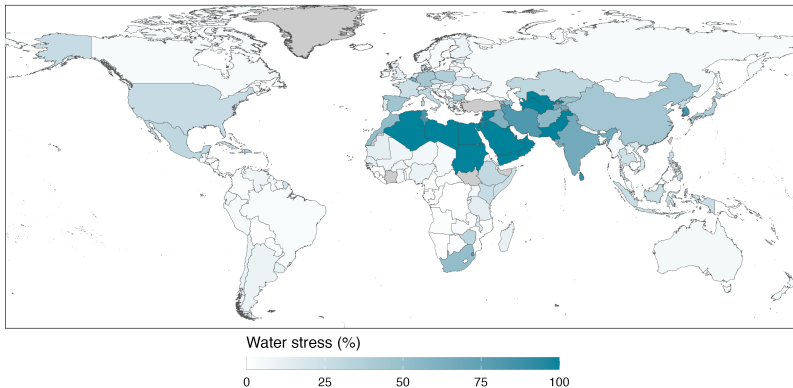
Fact 2: IV Robustness

[Back](#)

Water Stress Data: FAO [Back](#)

Definition (FAO AQUASTAT): Water stress = $\frac{\text{Total freshwater withdrawn}}{\text{Total renewable freshwater resources}} \times 100$

Fig. Water stress (%) across countries (FAO AQUASTAT, avg. 2008–2017)



Note: Values above 100% winsorized. Grey = missing data.

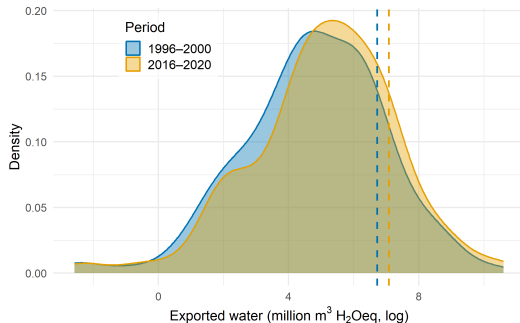
GLORIA MRIO: Details

[Back](#)

Data: GLORIA MRIO – 164 countries, 46 sectors, 1995–2020

Exported water content: water embedded in goods sold to foreign countries

- For each country i , sector k : water intensity ω_{ik} (m^3/USD) from GLORIA satellite accounts
- Exported water $\equiv \sum_{j \neq i} \sum_k \omega_{ik} \cdot X_{ij,k}$, where $X_{ij,k}$ = exports of k from i to j



Solving Algorithm: Static Equilibrium [Back](#)

Iterative algorithm for static GE given water stocks $\{R_i\}$ and tax rates $\{\tau_{i,k}\}$:

1. **Initialize:** guess $\{w_i, I_i\}$; normalize wages to numeraire w_{i_0}
2. **Prices:** compute $r_i = w_i G_i(R_i)$; unit costs $c_{i,k}$; cost shares $\alpha_{i,k}$; producer and consumer prices $P_{ji,k}, \tilde{P}_{ji,k}$
3. **Quantities:** compute $C_{ji,k}$; revenues $Y_{i,k}$; water use $H_{i,k} = (1 - \alpha_{i,k})Y_{i,k} / (r_i + \tau_{i,k})$; water labor $L_{i,0} = G_i(R_i) \sum_k H_{i,k}$; tax revenues T_i
4. **Update & check:** $w_i^{(new)} = \frac{\sum_k \alpha_{i,k} Y_{i,k}}{\bar{L}_i - L_{i,0}}$ (Labor MCC); $I_i^{(new)} = w_i \bar{L}_i + T_i$

→ Stop if $\max_i |(w_i^{(new)} - w_i) / w_i|$ and $\max_i |(I_i^{(new)} - I_i) / I_i|$ sufficiently small; else dampen and return to Step 1

Solving Algorithm: Dynamic & Steady State [Back](#)

Dynamic equilibrium: solve static GE each period t ; update stocks via the LOM:

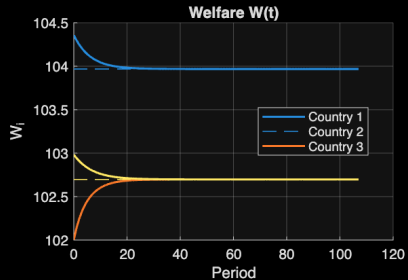
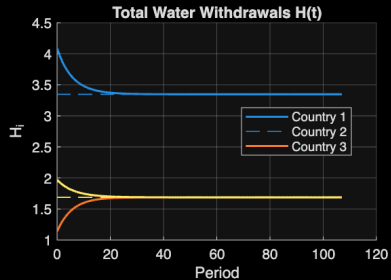
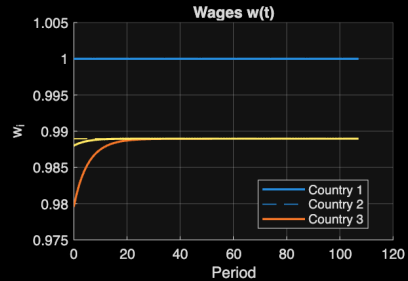
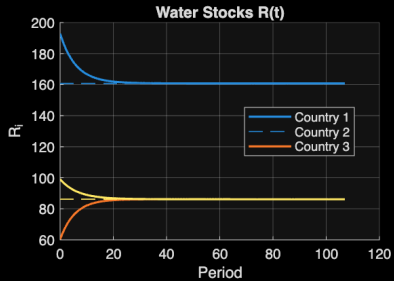
$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t) + \mathbf{\Psi}(t) \mathbf{H}(t)$$

Steady-state equilibrium: fixed-point intersection of two mappings

- (a) Static GE $\rightarrow \mathbf{H}$ as a function of $\mathbf{R}_B^{(ss)}$
- (b) LOM fixed point: $\mathbf{R}_B^{(ss)} = (\mathbf{I} - \mathbf{\Phi})^{-1} (\mathbf{\Gamma} + \mathbf{\Psi}^{(ss)} \mathbf{H}(\mathbf{R}_B^{(ss)}))$

$\Rightarrow$ Iterate on $\mathbf{R}_B^{(ss)}$ until both mappings are mutually consistent

Dynamic Equilibrium: Time Paths (dashed = steady state)



Optimal Policy: Dual Problem [Back](#)

Dual reformulation: planner directly chooses consumer prices

$$\mathbb{P}_i \equiv \{\tilde{P}_{ii,k}, \tilde{P}_{in,k}, \tilde{P}_{ni,k}; \tilde{r}_{i,k}\}_{n \neq i,k}$$

Income function for country i :

$$I_i(\cdot) = w_i L_i + \underbrace{(\tilde{r}_i - r_i)^\top H_i}_{\text{water taxes}} + \underbrace{(\tilde{P}_i - P_i)^\top C_i}_{\text{tariffs + prod. taxes}} + \underbrace{(\tilde{P}_{-i} - P_{-i})^\top C_{-i}}_{\text{export taxes}}$$

Key functions in the dual:

- Producer price: $P_{ij,k}(w_i, \tilde{r}_{i,k}) = \bar{d}_{ij,k} c_{i,k}(w_i, \tilde{r}_{i,k})$
- Water cost: $r_i(w_i, R_i) = w_i G_i(R_i)$
- Steady-state stock: $R_i = Z_i + \sum_{j,k} \lambda_{ji,k} H_{j,k}$

Optimal Policy: Wage Neutrality [Back](#)

Lemma 1 (Local wage neutrality): $\frac{\partial I_i(\cdot)}{\partial w_i} = 0$

Proof: expand using Shephard's lemma + labor market clearing:

$$\bar{L}_i - \underbrace{G_i(R_i) \sum_k H_{i,k}}_{=L_{i,0}} - \underbrace{\sum_k \frac{\alpha_{i,k} P_{ii,k} Q_{i,k}}{w_i}}_{=\sum_k L_{i,k}} = 0 \quad \checkmark$$

Wages only redistribute b/w wage payments and tax revenues when the planner controls all prices.

Assumption A1: policy-induced changes in *foreign* relative wages have no first-order effect on home income at the optimum.

Corollary 1: $\frac{\partial I_i(\cdot)}{\partial \mathbf{w}} = \mathbf{0} \Rightarrow$ wages drop from all FOCs

Optimal Policy: First-Order Conditions [Back](#)

FOC w.r.t. any policy instrument $\tilde{P}$ (after applying Corollary 1):

$$\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial \tilde{P}} + \sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} \left[\frac{\partial I_n}{\partial \tilde{P}} + \frac{\partial I_n}{\partial R} \frac{\partial R}{\partial \tilde{P}} + \frac{\partial I_n}{\partial H} \frac{\partial H}{\partial \tilde{P}} + \frac{\partial I_n}{\partial C} \frac{\partial C}{\partial \tilde{P}} \right] = 0$$

Key derivatives (1/2):

- **Direct price effect** (Roy's id.): $\frac{\partial V_n}{\partial \tilde{P}} = -\frac{\partial V_n}{\partial E_n} C_{in,k}$ if $\tilde{P} \in \tilde{P}_n$, else 0
- **Direct income, water:** $\frac{\partial I}{\partial \tilde{r}_{i,k}} = 0$ (water tax revenue and cost changes cancel exactly)

Key derivatives (2/2):

- **Income via water stocks:** $\frac{\partial I_i}{\partial R_i} = -\frac{G'_i(\cdot)}{G_i(\cdot)} r_i \sum_k H_{i,k}$
- **Income via water use:** $\frac{\partial I_i}{\partial H_{i,k}} = \tilde{r}_{i,k} - r_i$
- **Policy impact on stocks:** $\frac{\partial R_i}{\partial \tilde{P}} = \sum_{j,k} \lambda_{ji,k} \frac{\partial H_{j,k}}{\partial \tilde{P}}$

Optimal Policy: Decomposition of FOCs [Back](#)

FOCs decompose into three independent sub-problems:

$$\left\{ \begin{array}{ll} \sum_n \left(\frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} \pi_n \right) = \frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial V_i}{\partial E_i} & \text{(distributional objective)} \\ \tilde{\mathbf{P}} = \mathbf{P} & \text{(efficient non-water prices)} \\ \tilde{r}_{i,k} - r_i = \sum_n \frac{G'_n}{G_n} r_n \sum_k H_{n,k} \sum_g \lambda_{in,g} & \text{(efficient water prices)} \end{array} \right.$$

Complex GE derivatives $\partial \mathbf{C} / \partial \tilde{\mathbf{P}}$, $\partial \mathbf{H} / \partial \tilde{\mathbf{P}}$ cancel in both the goods and water FOCs.

Solutions to the three sub-problems:

- $\pi_i^* = \frac{(\partial \mathcal{W} / \partial W_i)(\partial V_i / \partial E_i) I_i}{\sum_n (\partial \mathcal{W} / \partial W_n)(\partial V_n / \partial E_n) I_n} \Rightarrow \Delta_i^* = \pi_i^* \sum_n I_n - I_i$
- $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$
- $\tau_{j,g} = \sum_i \frac{G'_i(\cdot)}{G_i(\cdot)} r_i \sum_k \lambda_{ji,g} H_{i,k}$